EXPERIMENT NO.10

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Aim: Explore the Cryptocurrency Landscape

Theory:

This experiment explores the cryptocurrency landscape by combining real-world market data analysis with blockchain technology research. The focus asset for this analysis is **Chainlink (LINK)**, and data was collected from three reputable sources as per the lab requirement:

- **Financial Data (APIs)** – Services like **CoinGecko** and **CoinMarketCap** provided structured data on market capitalization, circulating supply, and historical 24-hour trading volumes, which are essential for calculating liquidity and asset dominance.
- **News & Social Platforms** – Platforms such as **CryptoPanic** (news aggregator) and **LunarCrush** (social sentiment) were used to track how public perception and breaking news correlate with price volatility.
- **Regulatory & Institutional Portals** – Websites like **The Block Pro** or official government portals (e.g., SEC, EU MiCA updates) provided data on legal frameworks and institutional entry.

Cryptocurrency Landscape

- **Definition** – Cryptocurrencies are decentralized digital assets built on blockchain networks, enabling secure, transparent, and immutable transactions without central authority.
- **Market Diversity** – Thousands of cryptocurrencies exist, each serving unique roles:
    - **Store of Value (SoV)** – Bitcoin (BTC), characterized by digital scarcity.
    - **Utility Tokens** – Chainlink (LINK), Ether (ETH), used to pay for network services.

    - **Stablecoins** – USDT, USDC, pegged to fiat (USD) to provide price stability.
    - **Governance Tokens** – UNI, AAVE, granting holders voting rights over protocols.
- **Volatility** – Prices are highly sensitive to market sentiment, technological developments, and "macro" regulatory drivers.
- **Data Analysis Importance** – Integrating financial, social, and regulatory data is essential for a holistic understanding of asset performance and market health.

Data Collection Process

The program used the CoinGecko API to automate historical data fetching for Chainlink. The retrieved data was processed with Pandas to compute:

- **Daily Returns** – Percentage change between consecutive days to measure volatility.
- **Moving Averages (MA7, MA30)** – Short-term and long-term trend indicators.
- **Cumulative Returns** – Growth factor over time.
- **Rolling Volatility** – Short-term risk measurement based on daily return fluctuations.

Visualization & Dashboard

Using Matplotlib, a six-panel dashboard was created to display:

- **Price Trend** – Daily price movement over the selected period.
- **Price vs Volume** – Correlation between trading activity and price changes.
- **Daily Returns Distribution** – Statistical spread of daily gains/losses.
- **Moving Averages** – Technical analysis trend lines for price movement.
- **Cumulative Returns** – Overall growth performance.
- **Rolling Volatility** – Market risk trends over time.

Technological Analysis

Beyond market data, the study covered advancements in blockchain technology, including:

- **Consensus Mechanisms** – The shift toward Proof of Stake (PoS) for sustainability and Proof of History (PoH) for faster processing.
- **Scalability Solutions** – Implementation of Layer-2 Rollups (Optimistic and ZK-Rollups) and Sharding to solve the "Blockchain Trilemma."
- **Interoperability Protocols** – Tools like Chainlink CCIP and Polkadot's Relay Chain enabling seamless cross-chain data and value exchange.
- **Privacy Features** – Zero-Knowledge Proofs (ZKP) allowing verifiable transactions without revealing sensitive underlying data.
- **Technical Roadmaps** – Ethereum's focus on "The Surge" (100k+ TPS) and Solana's "Firedancer" client for institutional-grade reliability.

Market Trends & Adoption

- **Merchant Acceptance** – Significant surge in interest with major processors like Stripe and Shopify integrating stablecoins for global settlement.
- **Wallet Growth** – Increasing downloads of non-custodial wallets (MetaMask, Phantom), indicating a shift toward direct on-chain interaction.
- **Network Activity** – Analysis of transaction volumes vs. active addresses to distinguish between "whale" activity and organic retail growth.
- **Integration with Traditional Finance** – The rise of Real-World Asset (RWA) Tokenization, bringing traditional assets like bonds and real estate on-chain.
- **Regulatory Developments** – The implementation of frameworks like the EU's MiCA and U.S. FIT21 Act, providing a clear licensing roadmap for the industry

Steps:

1. **Install Required Libraries**
   Install Python libraries such as `requests`, `pandas`, and `matplotlib` using pip to enable data fetching, processing, and visualization
2. **Obtain API Key**
   Register on the CryptoCompare website and generate an API key to access cryptocurrency market data.
3. **Set Configuration Parameters**
   Define the required parameters in the code:
   - Cryptocurrency symbol (e.g., LINK)
   - Comparison currency (e.g., USD)
   - Number of days for historical data
4. **Fetch Cryptocurrency Data**
   Use the API to retrieve historical market data including price and trading volume.
5. **Create DataFrame**
   Convert the fetched data into a Pandas DataFrame and format timestamps into readable date-time values.
6. **Process the Data**
   Perform calculations to analyze the data:
   - Daily Returns
   - Moving Averages (7-day and 30-day)
   - Cumulative Returns
   - Rolling Volatility
7. **Generate Visualization Dashboard**
   Use Matplotlib to create multiple graphs:
   - Price Trend
   - Price vs Volume
   - Returns Distribution
   - Moving Averages
   - Cumulative Returns
   - Volatility
8. **Execute the Program**
   Run the Python script to generate the analysis and visual outputs.
9. **Analyze the Results**
   Interpret the graphs to understand price trends, market behavior, and risk levels.
10. **Draw Conclusion**
    Summarize insights about cryptocurrency performance and the importance of data analysis in understanding the crypto market.

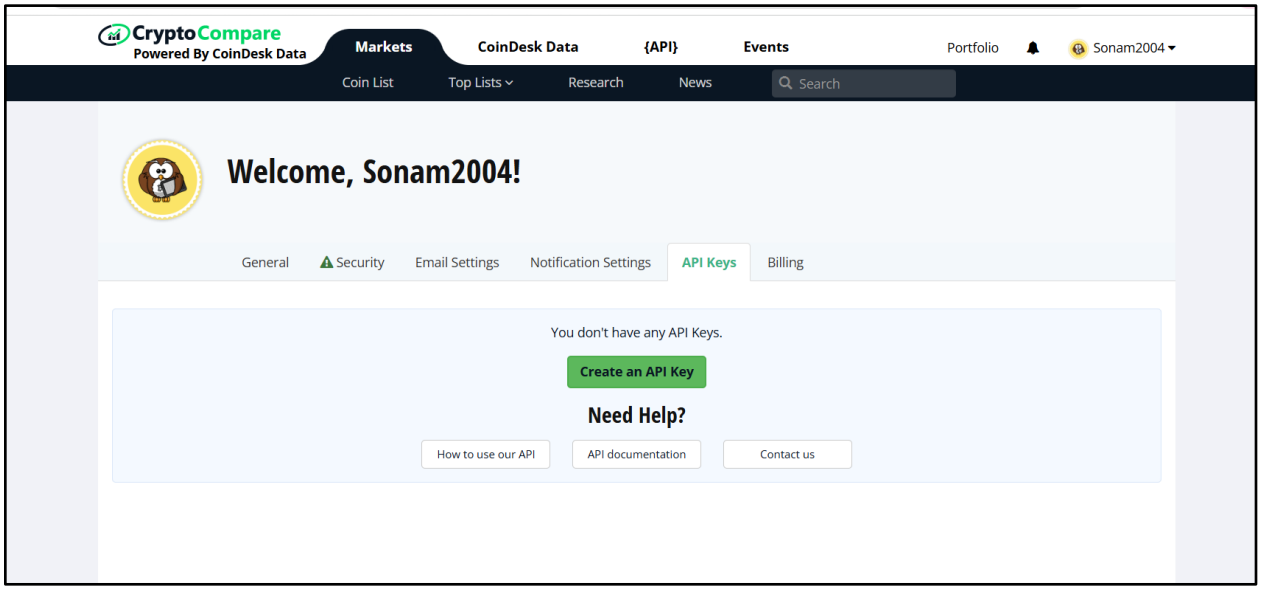
CODE:
Utilizing Api for analysis and visualization :
Code:

```
import requests
import matplotlib.pyplot as plt import
pandas as pd
import datetime

# ---------- CONFIGURATION -------
# Replace 'YOUR_API_KEY' with the key you got from CryptoCompare
API_KEY = 'YOUR_API_KEY'
SYMBOL = 'LINK'
COMPARISON_SYMBOL =
'USD'
LIMIT = 180    # Number of days

# ---------- Fetch Data Function ----------
def fetch_cryptocompare_data(symbol, comparison_symbol, limit, api_key): url =
    f"https://min-api.cryptocompare.com/data/v2/histoday"
    params = {
        "fsym": symbol,
        "tsym": comparison_symbol,
        "limit": limit,
        "api_key": api_key
    }

    response = requests.get(url, params=params) data =
    response.json()

    if data['Response'] == 'Success':
        # CryptoCompare returns OHLCV data in 'Data' -> 'Data' return
        data['Data']['Data']

    else:
        print(f"Error fetching data: {data.get('Message')}") return None
```

```
# ---------- Dashboard Function ----------
def cryptocompare_dashboard(symbol, comparison_symbol, limit, api_key): raw_data =
    fetch_cryptocompare_data(symbol, comparison_symbol, limit,
api_key)
    if not raw_data: return

    # Create DataFrame
    df = pd.DataFrame(raw_data)

    # Convert 'time' (unix timestamp) to datetime df['timestamp']
    = pd.to_datetime(df['time'], unit='s') df.set_index('timestamp',
    inplace=True)

    # Standardize column names to match your previous logic #
    'close' is used as the daily price
    df['price'] = df['close']
    df['volume'] = df['volumeto'] # Total volume in USD (volumeto)

    # Calculate metrics
    df['returns'] = df['price'].pct_change() df['MA7'] =
    df['price'].rolling(window=7).mean()
    df['MA30'] = df['price'].rolling(window=30).mean() df['cumulative'] = (1 +
    df['returns']).cumprod() df['volatility'] =
    df['returns'].rolling(window=7).std() * 100

    # ---------- Plotting ----------
    fig, axes = plt.subplots(3, 2, figsize=(16, 12)) fig.suptitle(f'CryptoCompare:
    {symbol} Analysis Dashboard',
fontsize=16, fontweight='bold')

    # 1. Price Trend
    axes[0, 0].plot(df.index, df['price'], color='tab:blue', linewidth=2) axes[0,
    0].set_title('Historical Price Trend (USD)')
    axes[0, 0].grid(True, linestyle='--', alpha=0.7)
```

```
# 2. Price vs Volume
axes[0, 1].plot(df.index, df['price'], color='tab:blue', label='Price')
axes[0, 1].bar(df.index, df['volume'], color='gray', alpha=0.3, label='Volume (USD)')
axes[0, 1].set_title('Price vs Trading Volume') axes[0, 1].legend()

# 3. Daily Returns Distribution
axes[1, 0].hist(df['returns'].dropna(), bins=40, color='tab:purple', edgecolor='white')
axes[1, 0].set_title('Daily Returns Distribution')

# 4. Moving Averages
axes[1, 1].plot(df.index, df['price'], color='lightgray', alpha=0.5, label='Price')
axes[1, 1].plot(df.index, df['MA7'], color='tab:orange', label='7-Day MA')
axes[1, 1].plot(df.index, df['MA30'], color='tab:red', label='30-Day MA')
axes[1, 1].set_title('Trend Analysis (Moving Averages)') axes[1, 1].legend()

# 5. Cumulative Returns
axes[2, 0].fill_between(df.index, df['cumulative'], color='tab:green', alpha=0.2)
axes[2, 0].plot(df.index, df['cumulative'], color='tab:green') axes[2, 0].set_title('Cumulative Growth')

# 6. Volatility
axes[2, 1].plot(df.index, df['volatility'], color='tab:red') axes[2, 1].set_title('7-Day Rolling Volatility (%)')

plt.tight_layout(rect=[0, 0.03, 1, 0.95]) plt.show()
```

OUTPUT:


CryptoCompare
Powered By CoinDesk Data
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Portfolio
Sonam2004
Coin List
Top Lists
Research
News
Search
Welcome, Sonam2004!
General
Security
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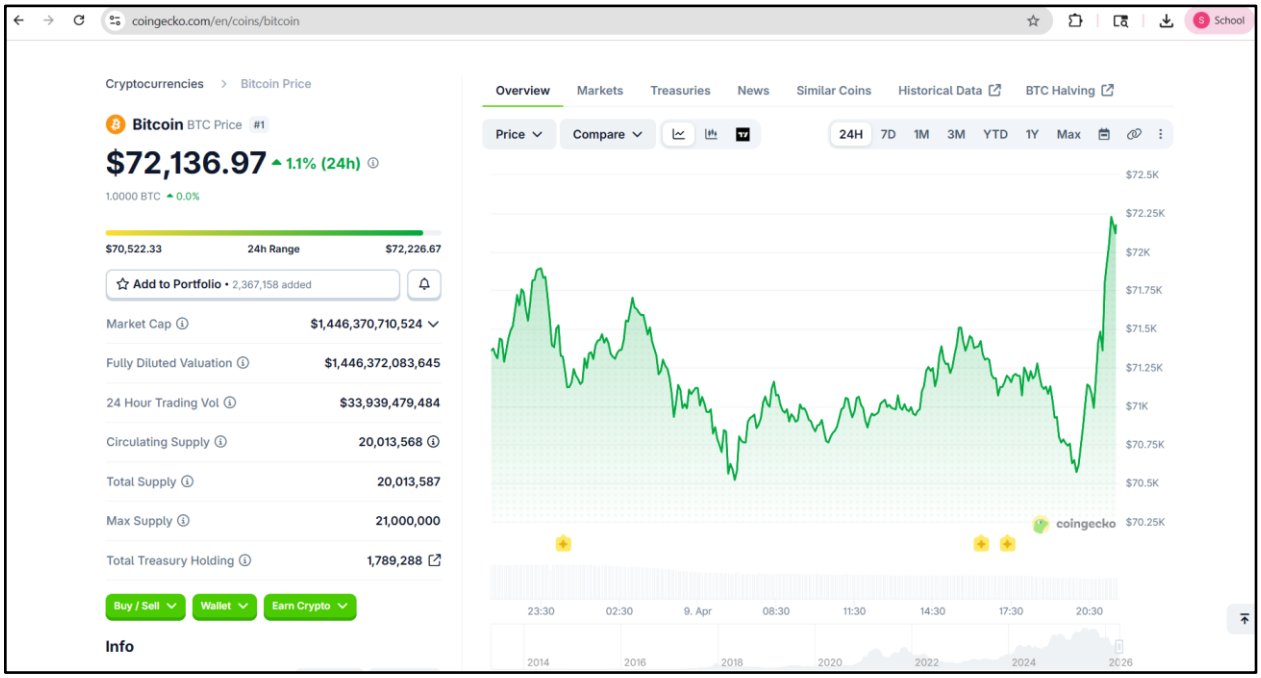
Notification Settings
API Keys
Billing
You don't have any API Keys.
Create an API Key
Need Help?
How to use our API
API documentation
Contact us


coingecko.com/en/coins/bitcoin
School
Cryptocurrencies
Bitcoin Price
Bitcoin BTC Price #1
$72,136.97 1.1% (24h)
1.0000 BTC 0.0%
$70,522.33
24h Range
$72,226.67
Add to Portfolio • 2,367,158 added
Market Cap $1,446,370,710,524
Fully Diluted Valuation $1,446,372,083,645
24 Hour Trading Vol $33,939,479,484
Circulating Supply 20,013,568
Total Supply 20,013,587
Max Supply 21,000,000
Total Treasury Holding 1,789,288
Buy / Sell
Wallet
Earn Crypto
Info
Overview
Markets
Treasuries
News
Similar Coins
Historical Data
BTC Halving
Price
Compare
24H
7D
1M
3M
YTD
1Y
Max
$72.5K
$72.25K
$72K
$71.75K
$71.5K
$71.25K
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17:30
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2014
2016
2018
2020
2022
2024
2026

Now LIVE: Spot CEX Report 2026

Coins: 17,728 Exchanges: 1,457 Market Cap: $2.533T ▲0.4% 24h Vol: $83.759B Dominance: BTC 57.1% ETH 10.6% Gas: 0.189 GWEI

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Cryptocurrency Prices by Market Cap

Highlights

The global cryptocurrency market cap today is $2.53 Trillion, a ▲0.4% change in the last 24 hours. Read more

$2,532,738,465,571
Market Cap ▲0.4%

$83,758,769,423
24h Trading Volume

Trending	View more >	Top Gainers	View more >
Aria.AI	$0.352 ▼44.1%	Xphere	$0.01688 ▲70.0%
Pudgy Penguins	$0.006571 ▼0.9%	RaveDAO	$0.454 ▲51.0%
Bitcoin	$72,168.56 ▲1.1%	Magma Finance	$0.1696 ▲32.7%

All | Highlights | Base Ecosystem | Categories | Terminal of Truths | Parallelized EVM | Quantum-Resistant | Customize

#	Coin		Price	1h	24h	7d	24h Volume	Market Cap	Last 7 Days
1	Bitcoin BTC	Buy	$72,168.56	▲1.9%	▲1.1%	▲8.3%	$34,128,837,885	$1,447,227,188,637	

All | Highlights | Base Ecosystem | Categories | Parallelized EVM | Quantum-Resistant | Entertainment | Customize

#	Coin		Price	1h	24h	7d	24h Volume	Market Cap	Last 7 Days
1	Bitcoin BTC	Buy	$72,277.98	▲1.7%	▲1.4%	▲8.0%	$33,964,623,535	$1,445,347,547,145	
2	Ethereum ETH	Buy	$2,222.68	▲2.1%	▲0.3%	▲7.5%	$15,146,771,229	$267,989,384,582	
3	Tether USDT		$1.00	▲0.0%	▲0.0%	▲0.0%	$55,449,851,675	$184,118,836,658	
4	XRP XRP	Buy	$1.35	▲1.7%	▼0.0%	▲4.2%	$2,408,868,945	$83,143,006,976	
5	BNB BNB	Buy	$606.76	▲1.0%	▲0.2%	▲5.3%	$906,312,248	$82,739,987,893	
6	USDC USDC		$1.00	▲0.0%	▲0.0%	▼0.0%	$12,014,578,070	$78,231,682,358	
7	Solana SOL	Buy	$83.80	▲2.1%	▲0.5%	▲5.9%	$2,842,320,938	$48,104,453,648	
8	TRON TRX	Buy	$0.3188	▲0.0%	▲0.4%	▲0.8%	$435,361,076	$30,202,185,677	

CoinMarketCap Cryptocurrencies Dashboards DexScan Exchanges Community Products Portfolio W

Top Trending Watchlist Prediction Markets Most Visited New More

Market Cap
$2.45T 0.9%

CMC20
$148.16 0.96%

Fear & Greed
46
Neutral

Altcoin Season
34/100
Bitcoin Altcoin

Topic ETH Foundation just dumped $8M more | New Top News | Add AI alert | BTC gets prototype quantum-resistant wallet

All Networks BSC Solana Base Ethereum More

#	Name		Price	1h %	24h %	7d %	Market Cap
	CoinMarketCap 20 Index DTF	CMC20 Buy	$147.18	1.58%	1.02%	7.22%	$15,182,079
1	Bitcoin BTC	Buy	$72,203.77	1.63%	1.13%	7.98%	$1,445,056,613,203
2	Ethereum ETH	Buy	$2,220.06	2.07%	0.16%	7.90%	$267,941,915,050

CoinMarketCap Cryptocurrencies Dashboards DexScan Exchanges Community Products Portfolio Watchlist Search

Bitcoin BTC #1 6M

$72,158.09 1.03% (24h)

Why is BTC's price down today?

Market cap
$1.44T 1.15%

Volume (24h)
$33.74B 37.28%

Vol/Mkt Cap (24h)
2.66%

FDV
$1.51T

Total supply
20.01M BTC

Max. supply
21M BTC

Circulating supply
20.01M BTC

Treasury Holdings
1.17M BTC

Profile score 100%

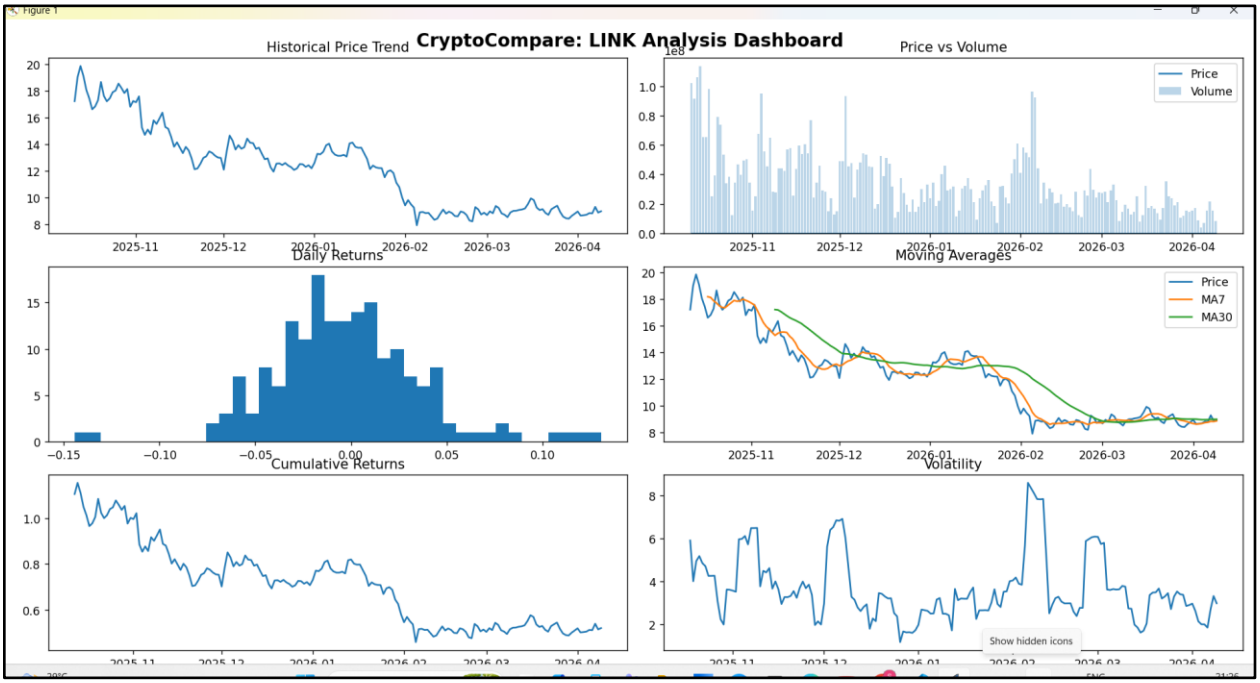
Boost

Website Website Whitepaper

Chart Markets News Yield Market Cycles Treasuries

Buy BTC

Price Mkt Cap TradingView Compare 24h 1W 1M 1Y All Log



Conclusion:-

By combining automated market data processing, visual trend analysis, blockchain technology evaluation, and adoption metrics, this experiment presents a holistic view of the cryptocurrency ecosystem. It demonstrates how blockchain's technical foundations connect with real-world market behavior, providing insights into both current trends and future advancements in the digital asset industry.